NEWTON SCHOOL

ALGEBRA 1

Pythagorean Theorem & Distance Formula — Review

Student: Evelyn Yoo **Date:** 2026-05-27

Standard: *CCSS.MATH.CONTENT.8.G.B.7 & 8.G.B.8 — Apply the Pythagorean Theorem to determine unknown side lengths in right triangles and to find the distance between two points in a coordinate plane.*

> **Before you start:** *Read each problem twice. If it says Find, you are* **solving** *— your answer is one number. Do not invent extra steps the problem did not ask for. Show the equation you used, then your work.*

Part A — Find the Missing Side

For each right triangle, use the Pythagorean Theorem $a^2 + b^2 = c^2$ where c is the hypotenuse. **Find** *the missing length. Leave answers as integers or in simplified radical form.*

1. Legs $a = 6$ and $b = 8$. Find the hypotenuse c.

Answer: $c =$ ___________

2. Legs $a = 5$ and $b = 12$. Find the hypotenuse c.

Answer: $c =$ ___________

3. One leg $a = 9$, hypotenuse $c = 15$. Find the other leg b.

Answer: $b =$ ___________

4. Hypotenuse $c = 25$, one leg $b = 24$. Find the other leg a.

Answer: $a =$ ___________

Part B — Pythagorean Word Problems

Draw *a right triangle for each situation. Label the two known sides and the unknown side. Then apply the theorem.*

5. A 13 ft ladder leans against the back wall of a comedy club. The base of the ladder is 5 ft away from the wall. How high up the wall does the ladder reach?

Answer: ___________ ft

6. A rectangular stage at the Just For Laughs Festival measures 9 ft by 12 ft. A microphone cable runs from one corner of the stage to the opposite corner along the diagonal. How long is the diagonal?

Answer: ___________ ft

Part C — Distance Formula

Use the Distance Formula $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ *to find the distance between the two points. Leave answers as exact values (integers or simplified radicals).*

7. Find the distance between $(1,\ 2)$ and $(4,\ 6)$.

Answer: $d =$ ___________

8. Find the distance between $(-3,\ 4)$ and $(2,\ -8)$.

Answer: $d =$ ___________

Part D — Mixed Challenge

These problems connect both ideas. Show every step.

9. On a map of Montreal, the comedy stage is at point $(2,\ 1)$ and the back exit is at point $(7,\ 13)$. Each unit on the map equals one meter. How far apart are the stage and the exit?

Answer: ___________ meters

10. Triangle $\triangle ABC$ has vertices $A(0,\ 0)$, $B(8,\ 0)$, and $C(8,\ 6)$.

(a) Use the Distance Formula to find the length of side AC.

(b) Verify your answer using the Pythagorean Theorem on the right triangle with legs AB and BC. Both methods should give the same length.

Answer: $AC =$ ___________